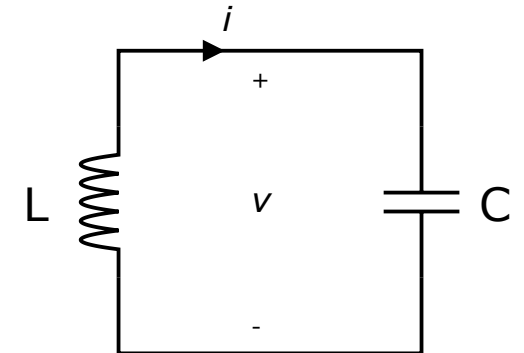
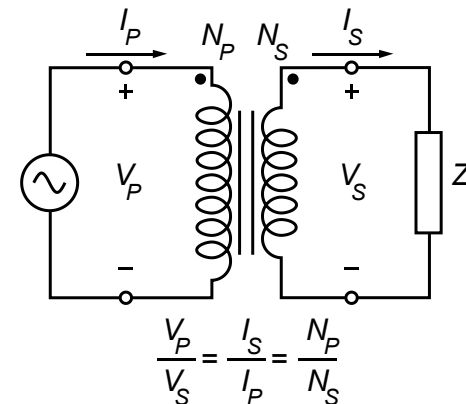
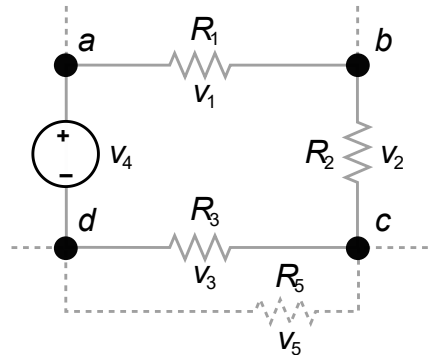
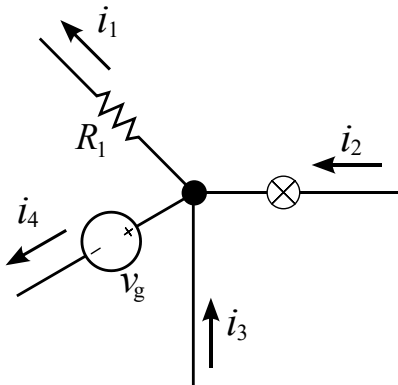


Introduction to Electrical Engineering



Overview

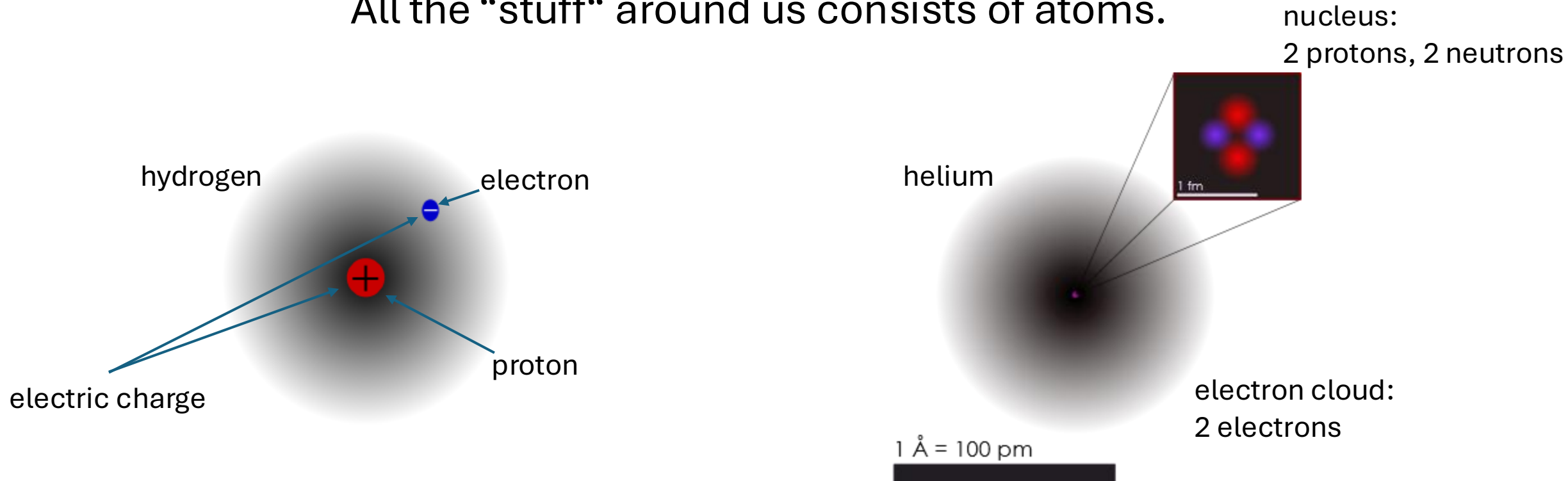
- 1. Physical Principles and Basic Concepts**
2. Direct Current Circuits
3. Electric and Magnetic Fields
4. Alternating Current Technology

Physical Principles and Basic Concepts

- Electric Charge
- Band Structure
- Current
- Voltage
- Ohm's Law
- Resistors
- Ideal Voltage & Current Sources
- Electric Power

The World Is Not What It Seems

All the “stuff” around us consists of atoms.



And atoms are nearly empty.

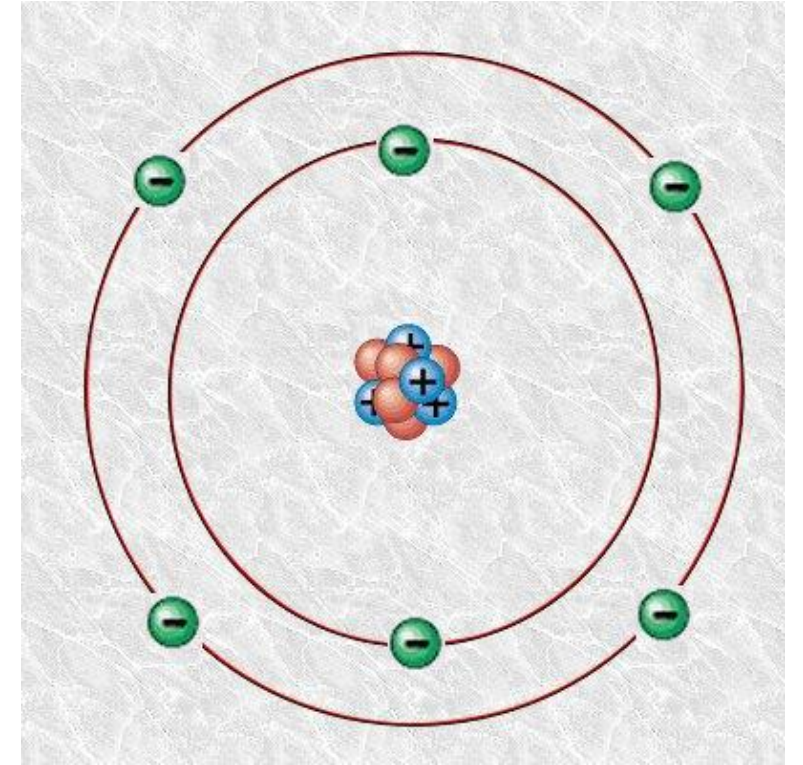
Periodic Table of Elements

Big Bang fusion Cosmic ray fission Dying low-mass stars Merging neutron stars Exploding massive stars Exploding white dwarfs Human synthesis No stable isotopes																		He 2					
H 1	Li 3	Be 4																B 5	C 6	N 7	O 8	F 9	Ne 10
Na 11	Mg 12																Al 13	Si 14	P 15	S 16	Cl 17	Ar 18	
K 19	Ca 20	Sc 21	Ti 22	V 23	Cr 24	Mn 25	Fe 26	Co 27	Ni 28	Cu 29	Zn 30	Ga 31	Ge 32	As 33	Se 34	Br 35	Kr 36						
Rb 37	Sr 38	Y 39	Zr 40	Nb 41	Mo 42	Tc 43	Ru 44	Rh 45	Pd 46	Ag 47	Cd 48	In 49	Sn 50	Sb 51	Te 52	I 53	Xe 54						
Cs 55	Ba 56			Hf 72	Ta 73	W 74	Re 75	Os 76	Ir 77	Pt 78	Au 79	Hg 80	Tl 81	Pb 82	Bi 83	Po 84	At 85	Rn 86					
Fr 87	Ra 88			La 57	Ce 58	Pr 59	Nd 60	Pm 61	Sm 62	Eu 63	Gd 64	Tb 65	Dy 66	Ho 67	Er 68	Tm 69	Yb 70	Lu 71					
		Ac 89	Th 90	Pa 91	U 92	Np 93	Pu 94	Am 95	Cm 96	Bk 97	Cf 98	Es 99	Fm 100	Md 101	No 102	Lr 103							

Coulomb Force

$$F = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{r^2}$$

$$\epsilon_0 = 8,854 \cdot 10^{-12} \frac{\text{As}}{\text{Vm}}$$



positive and negative charges attract each other

simple atomic model: balance with centrifugal force

→ stable electron orbits

Electric Charge

$$Q = n \cdot e$$

elementary charge: $e = 1,6021892 \cdot 10^{-19} \text{ C}$

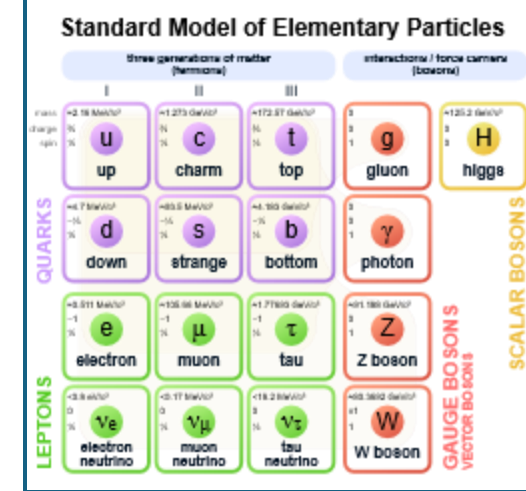
$$[Q] = 1 \text{ C} \quad (\text{coulomb})$$

e : fundamental physical property

(odd number because some physical units were arbitrarily determined)

other fundamental forces: gravity (mass, only one not quantized?), strong (keeps atomic nuclei together) and weak (e.g., reason for radioactive beta decay) nuclear forces

not relevant for electrical engineering though: gravity very weak, nuclear forces very short-ranged

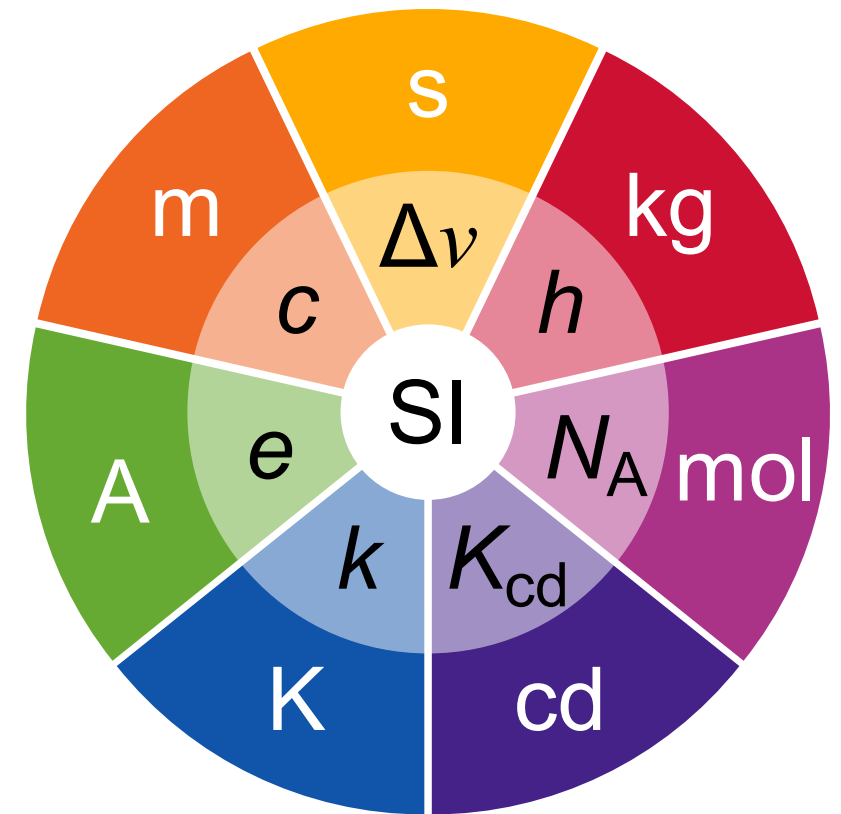
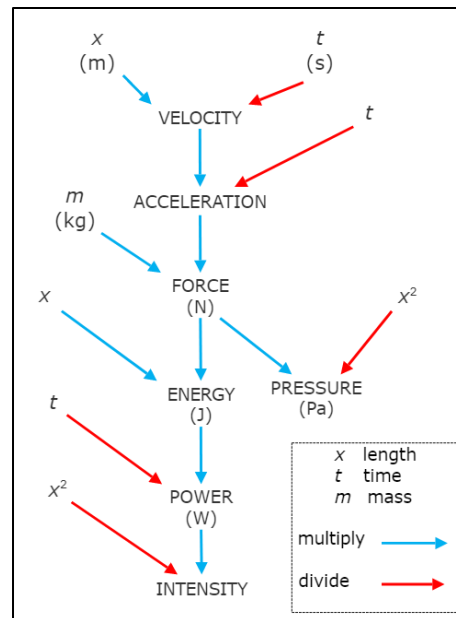


International Systems of Units (SI)

just a convention

- time
- length
- mass
- electric current
- thermodynamic temperature
- amount of substance
- luminous intensity

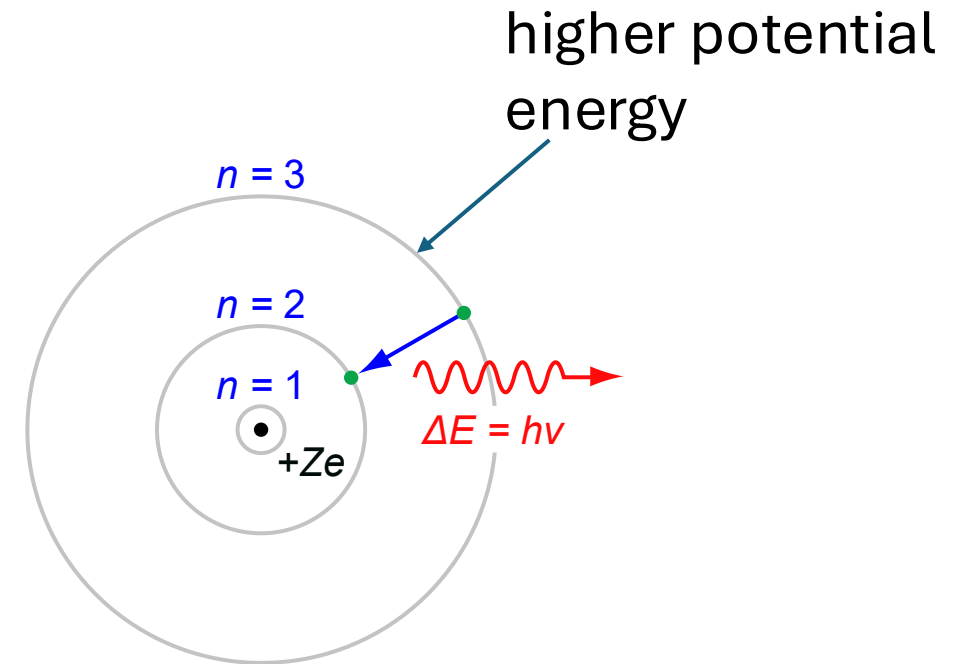
all other units derived



outer ring: SI base units
inner ring: constants

Bohr Model

allowing only certain energies (orbits) of electrons → quantized!



modern view: orbital model (location of electrons in atom determined by quantum-mechanical wave function → probability distribution)

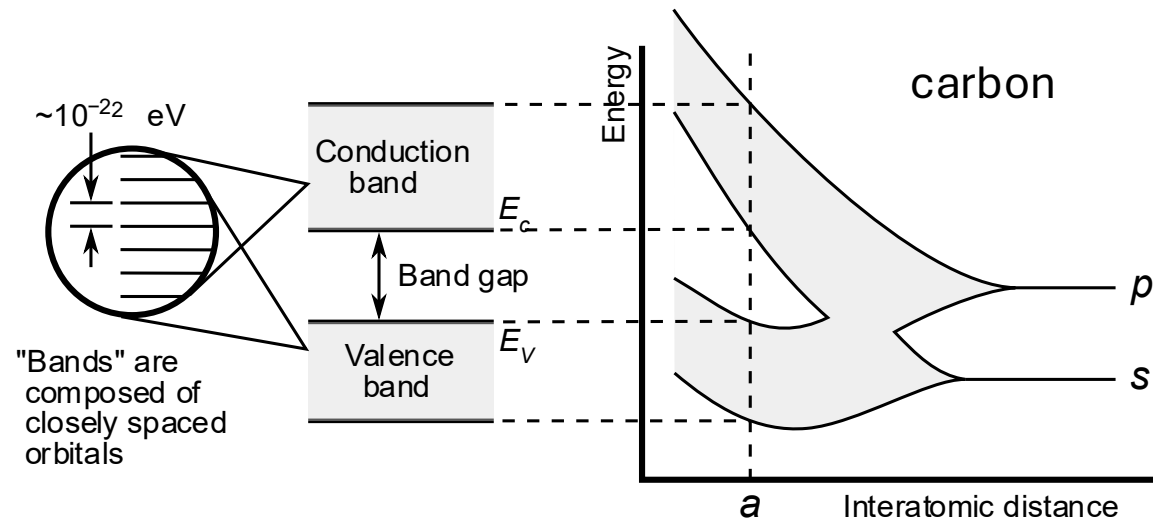
models just try to approximate the world: more complex models capture more details

Band Structure

Atomic nuclei have fixed positions in solids.

But the electron shells overlap.

→ Higher-energetic electrons can move (drift) through solid matter.



Electric Current

moving electrons → moving charges: electric current

$$I = \frac{Q}{t}$$

$$i(t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta Q}{\Delta t} = \frac{dQ}{dt}$$

$$[I] = 1 \text{ A} \quad (\text{ampere})$$

Drift Velocity

$$Q = e \cdot n \cdot A \cdot l$$

$$\rightarrow v_{\text{drift}} = \frac{I}{enA}$$

typically, 1.5 mm² copper cable
used for lighting and 2.5 mm² for
power circuits

typically, 16 A fuses
in households

example:

copper ($n = 8,49 \cdot 10^{19} \text{ }^1/\text{mm}^3$) wire of $A = 1.5 \text{ mm}^2$ with $I = 15 \text{ A}$

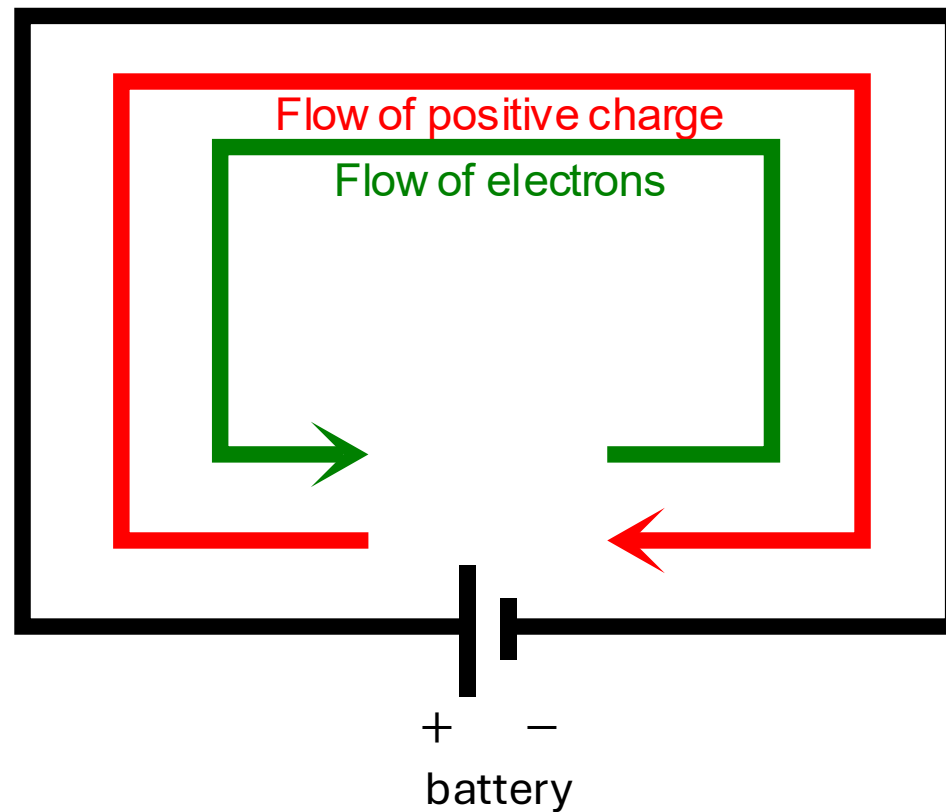
$$\rightarrow v_{\text{drift}} = 0.74 \frac{\text{mm}}{\text{s}}$$

but starts everywhere in the conductor at the same time

$J = \frac{I}{A}$ is current density (measure of thermal stress on conductor)

Conventional Current

arbitrarily defined as direction in which positive charges flow



Voltage

Voltage is the driving force of electric current.

relative quantity: potential difference between two points in space

$$V = V_{12} = \varphi(P_1) - \varphi(P_2)$$

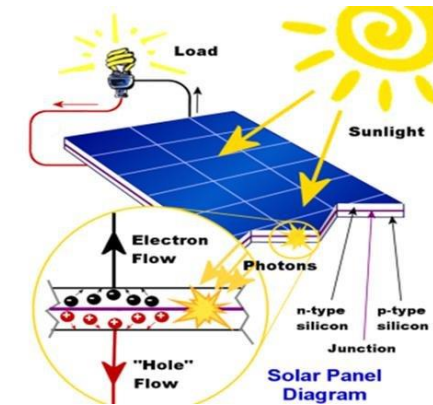
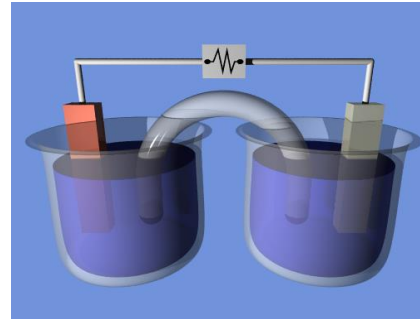
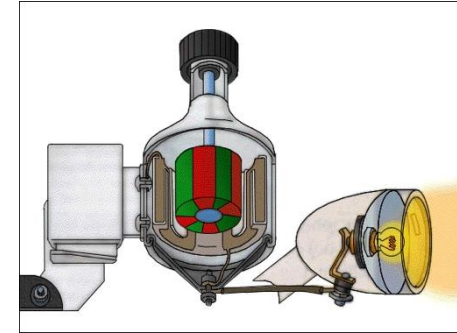
symbol U in German literature

electric potential $\varphi = \frac{W}{Q}$: defined from potential energy of a charge

$$[V] = [\varphi] = 1 \text{ V} \quad (\text{volt})$$

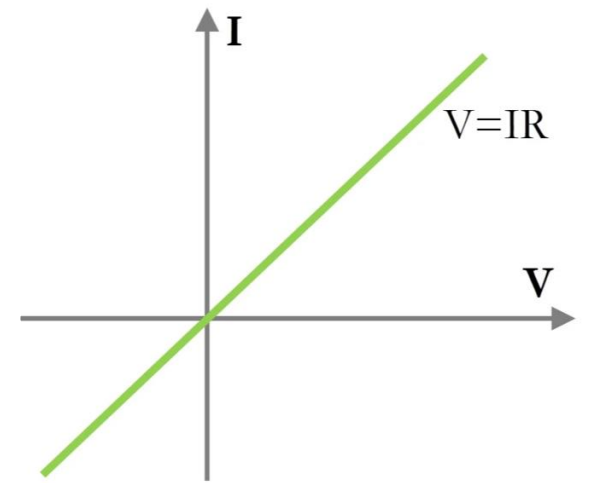
Types of Voltage Generation

- generators (mechanical, often via turbines, e.g., from thermal energy)
- batteries (chemical)
- photovoltaic
- piezoelectric
- ...



Ohm's Law

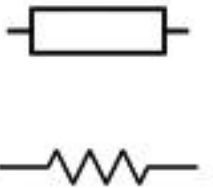
$$V = R \cdot I$$



current-voltage characteristic
(I-V curve)

$$[R] = 1 \, \Omega \quad (\text{ohm})$$

electrical resistance



Electrical Conductance

$$G = \frac{1}{R} = \frac{I}{V}$$

$$[G] = 1 \text{ S} \quad (\text{siemens})$$

formerly: $[G] = 1 \text{ } \Omega^{-1} \text{ (mho)}$

Specific Resistance

at 20 °C

$$R = \varrho \frac{l}{A}$$

$$[\varrho] = 1 \frac{\Omega \text{mm}^2}{\text{m}}$$

also called electrical resistivity

example:

copper cable with $A = 2.5 \text{ mm}^2$ and $l = 10 \text{ m}$ has $R = 0.07 \Omega$

→ wiring usually negligible

material	ϱ ($\Omega \text{mm}^2/\text{m}$)
silver	0.016
copper (power cable)	0.017 ... 0.018
gold	0.022
aluminium	0.027
constantan (Cu + Ni)	0.5
quartz glass	$7.5 \cdot 10^{23}$

Specific Conductance

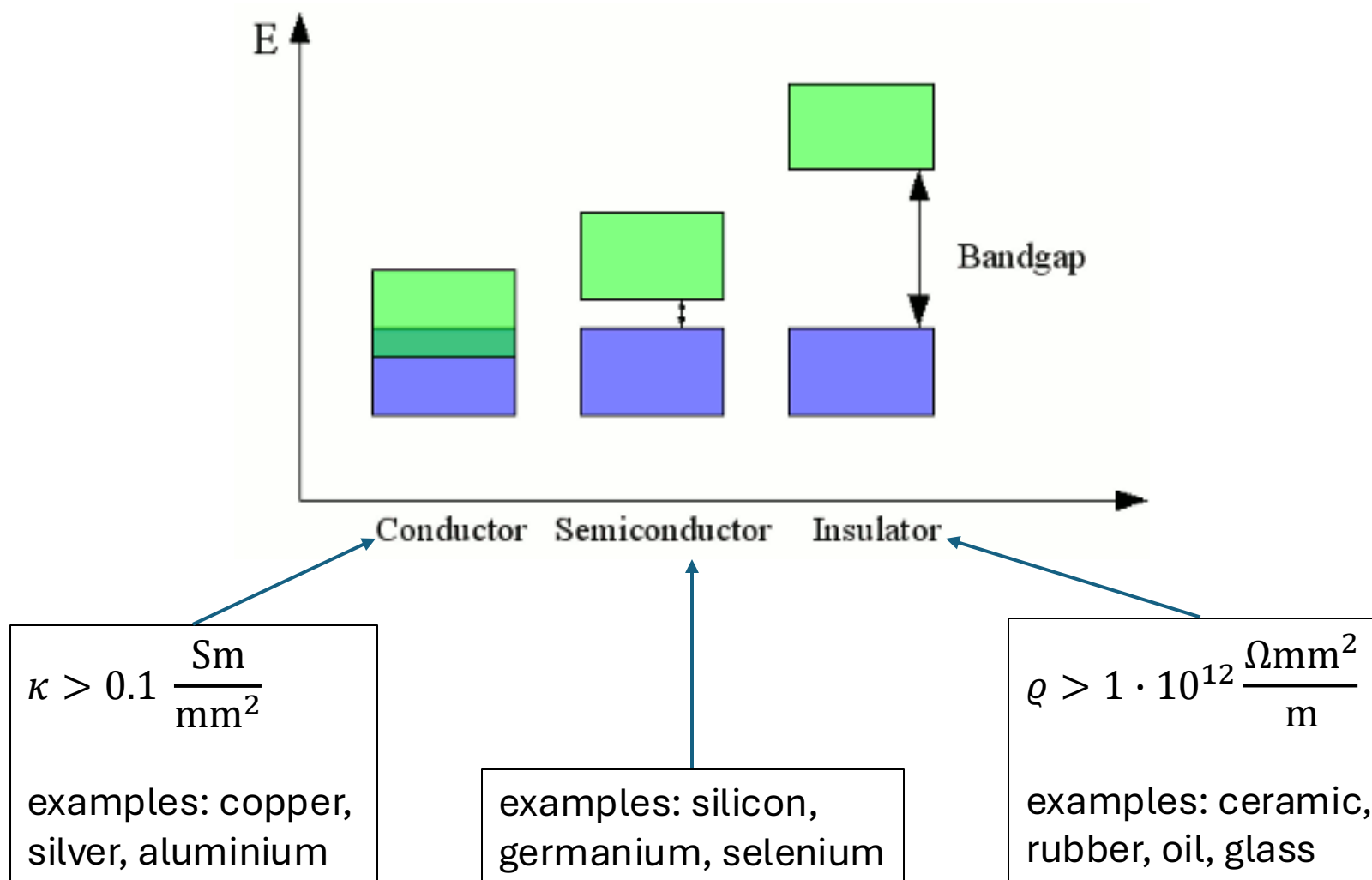
$$\kappa = \frac{1}{\varrho}$$

$$[\kappa] = 1 \frac{\text{Sm}}{\text{mm}^2}$$

$$G = \kappa \frac{A}{l}$$

also called electrical conductivity

Conductor, Semiconductor, Insulator



Temperature Dependence of Resistivity

- metallic conductors: resistivity increases with higher temperature (stronger vibrations of atoms, impeding the flow of electrons)
- semiconductors: resistivity decreases with higher temperature (more electrons can jump to the conduction band, increasing the number of free charge carriers, i.e., electrons and holes)
- insulators: decreases slightly with higher temperature

Temperature Coefficient

for $T_0 = 20\text{ }^{\circ}\text{C}$

$$\varrho(T) = \varrho(T_0) \left(1 + \alpha_{T_0} (T - T_0) \right)$$

$$[\alpha] = \frac{1}{\text{K}}$$

NB: linear approximation only valid in limited temperature range

metals: positive temperature coefficient (PTC)

semiconductors: negative temperature coefficient (NTC)

material	α_{20} in K^{-1}
iron	0.0065
aluminium	0.0043
gold	0.0034
copper	0.0039
silver	0.0038
constantan	0.00003
germanium	-0.05
silicon	-0.07

Example: Tungsten Filament

A light bulb contains a tungsten filament ($\kappa_{20} = 18.2 \text{ Sm/mm}^2$, $\alpha_{20} = 4.1 \cdot 10^{-3} \text{ K}^{-1}$, $\beta_{20} = 1 \cdot 10^{-6} \text{ K}^{-2}$) with a diameter of $24 \text{ }\mu\text{m}$ and a length of 62 cm , which is operated at a constant voltage of 230 V .

Current through the tungsten wire when the bulb is switched on in cold state ($20 \text{ }^\circ\text{C}$):

$$I_{20} = \frac{V}{R_{20}} = \frac{V}{\frac{l}{\kappa_{20}A}} = 3.05 \text{ A}$$

at operating temperature ($2200 \text{ }^\circ\text{C}$):

$$I_{2200} = \frac{V}{R_{2200}} = \frac{V}{R_{20}[1 + \alpha_{20}(2200 - 20) + \underbrace{\beta_{20}(2200 - 20)^2}_{\text{quadratic term: more accurate approximation for large temperature differences}}]} = 0.21 \text{ A}$$

quadratic term: more accurate approximation
for large temperature differences

Electrical Loads

any component/device consuming electrical power

resistive load:

electrical energy is converted into heat or light (e.g., incandescent bulbs)

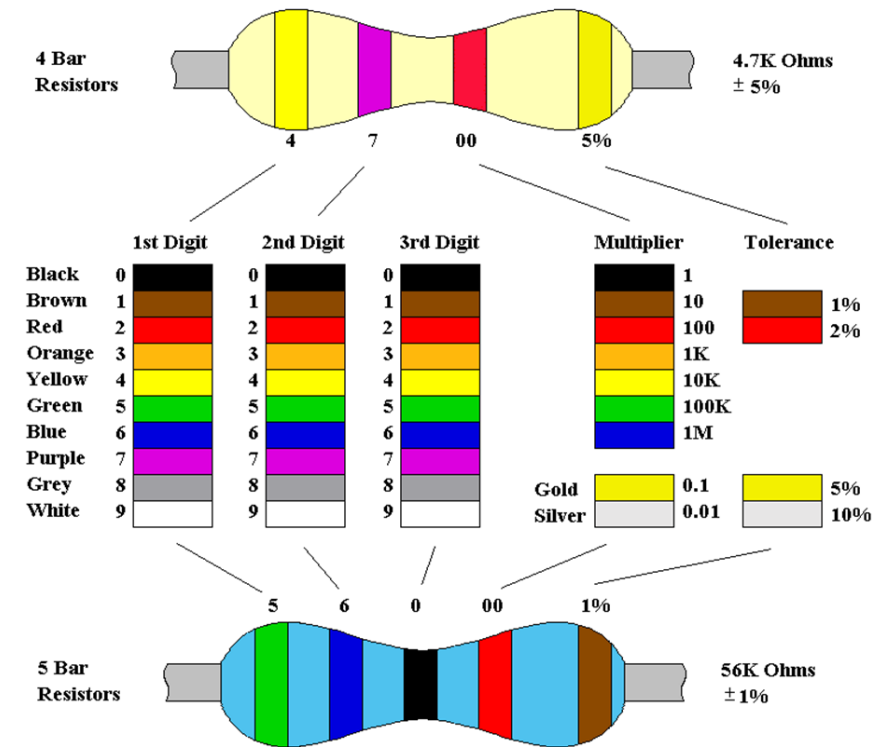
resistor (also a resistive load itself):

specific component to limit/control the current in electrical circuits
(precise, known amount of resistance)

capacitive & inductive loads: energy temporarily stored in electric or magnetic fields

Resistors

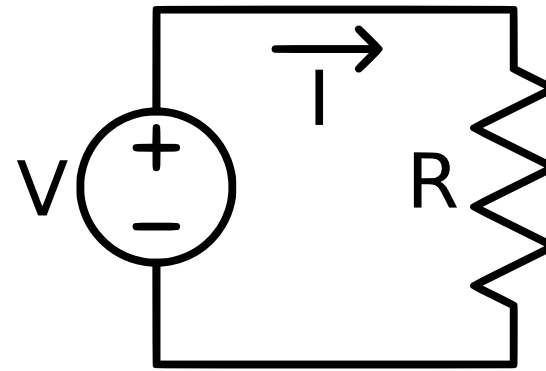
E6 (±20%)	E12 (±10%)	E24 (±5%)	E6 (±20%)	E12 (±10%)	E24 (±5%)
1,0	1,0	1,0	3,3	3,3	3,3
		1,1			3,6
		1,2			3,9
1,5	1,5	1,3	4,7	4,7	4,3
		1,6			5,1
		1,8			5,6
2,2	2,2	2,0	6,8	6,8	6,2
		2,2			7,5
		2,4			8,2
2,7	2,7	2,7	8,2	8,2	8,2
		3,0			9,1



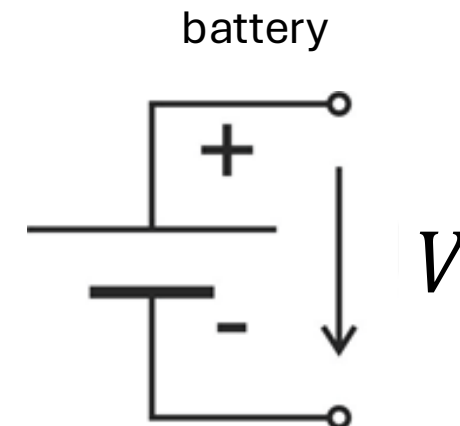
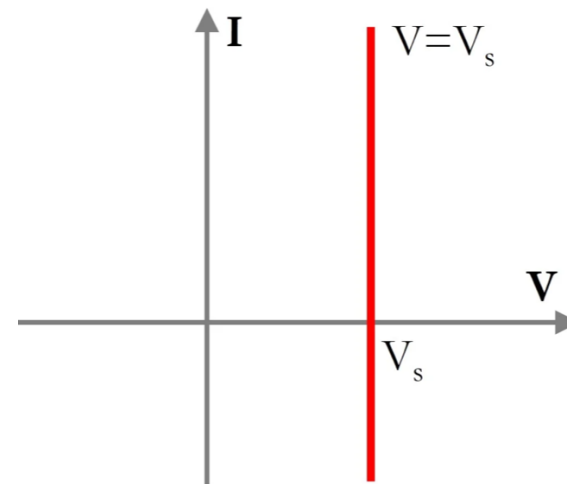
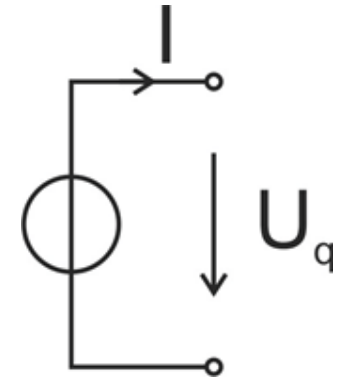
Ideal Voltage Source

model!

maintains a fixed voltage
across its two terminals
(independent of the output
current or load resistance)



also:

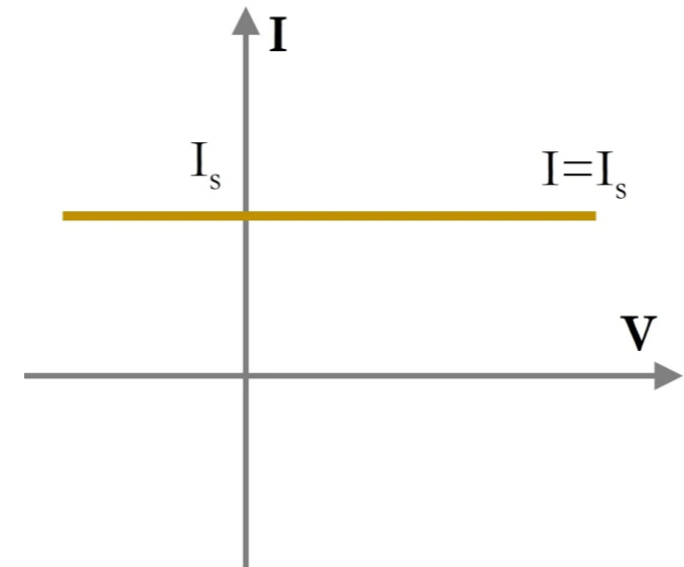
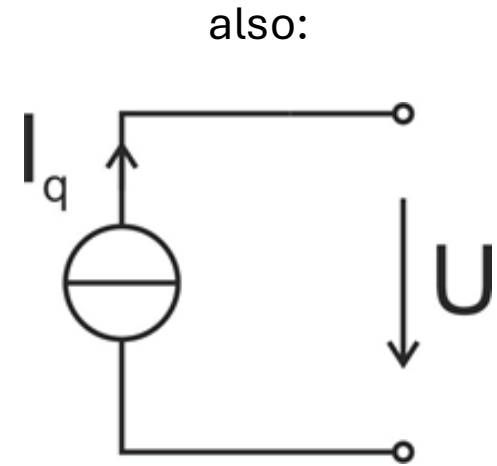
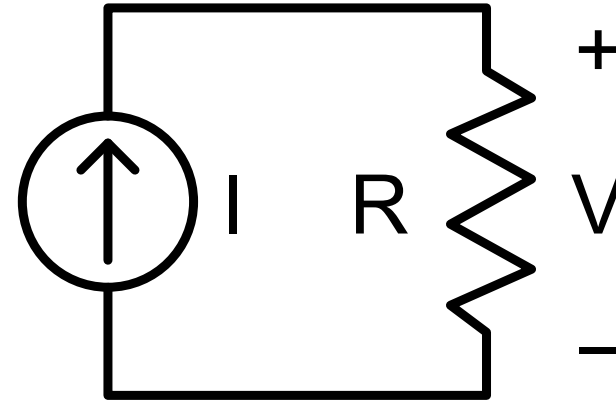


Ideal Current Source

generates a fixed current
(independent of the voltage
changes across it)

Usually, there are only voltage
supplies.

However, the behavior of the
current source model can be
constructed using circuitry.



Work

electrical energy dissipated in a resistor:

$$W = V \cdot Q = V \cdot I \cdot t$$

if current or voltage varies over time:

$$W = \int_0^t V \cdot I \, dt$$

Electric Power

$$P = \frac{W}{t} = V \cdot I$$

$$[P] = 1 \text{ W (watt)}$$

example:

A cooler bag for connection to the 12 V electrical system of a car draws a constant current of 4 A, resulting in $P = 48 \text{ W}$. So, in 8 hours $W = 0.384 \text{ kWh}$.

Summary

A basic **electrical circuit** consists of

- a **voltage source** (providing the electrical potential difference that drives the current, like a battery or generator),
- **conductive pathways** (wires or traces allowing the current to flow from the source to the load and back),
- and a **load** (device using the electrical energy, such as a light bulb, motor, or computer).

Ohm's law:
$$I = \frac{V}{R}$$